1 Planning (15 marks)

Defining the problem (3 marks)

P *f* is the independent variable or vary *f.* [1]

P *h* is the dependent variable or measure *h.* [1]

P Keep current in coil constant. [1]

Methods of data collection (5 marks)

M Labelled diagram of apparatus: including working circuit with a.c. supply, coil, stand and ring. [1]

M Signal generator (or variable frequency power generator). [1]

M Measure *h* with a rule/calipers. [1]

M Measure *f* using oscilloscope/read off signal generator. [1]

M Measure *h* from opposite sides of the ring (and determine mean)/wait for ring to stabilise. [1]

Method of analysis (2 marks)

A Plot a graph of lg *h* against lg *f*. (Allow ln – ln graphs.) [1]

A $k = 10^{y\text{-intercept}}$ and n = gradient. [1]

Safety considerations (1 mark)

S Reasoned method to prevent the coil overheating – switch off when not in use. Or reasoned method to prevent injury from hot coil – do not touch hot coil, use gloves. Do not allow "small currents". [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Use iron/steel retort stand
2 Method to keep current constant
3 Method to determine period from oscilloscope using time-base
4 When using oscilloscope, $f = 1/T$
5 Use coil of many turns/large current to have measurable heights
6 Logarithmic equation e.g. $\lg h = n \lg f + \lg k$
7 Relationship is valid if a straight line is produced

Do not allow vague computer methods.

[Total: 15]

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